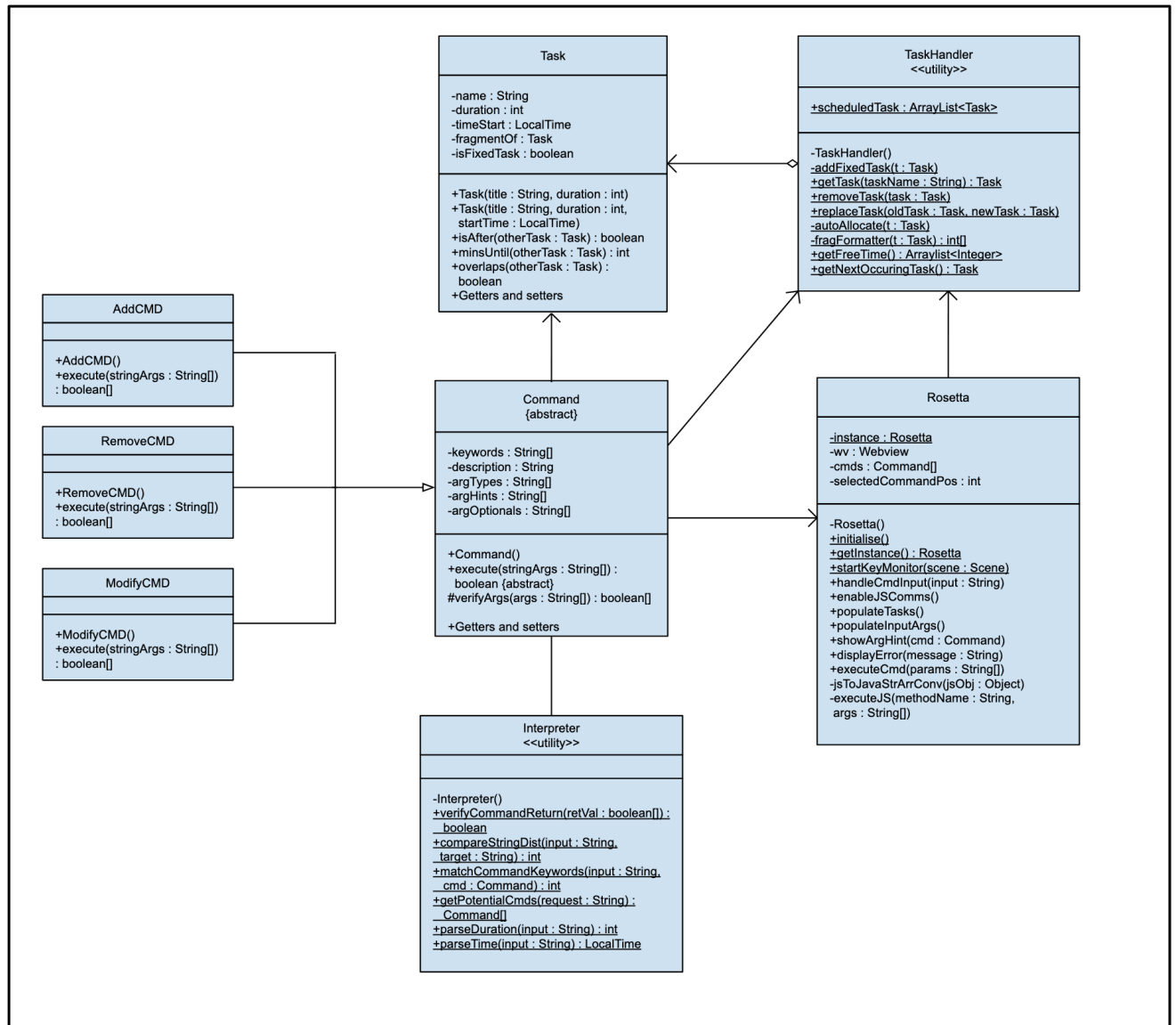
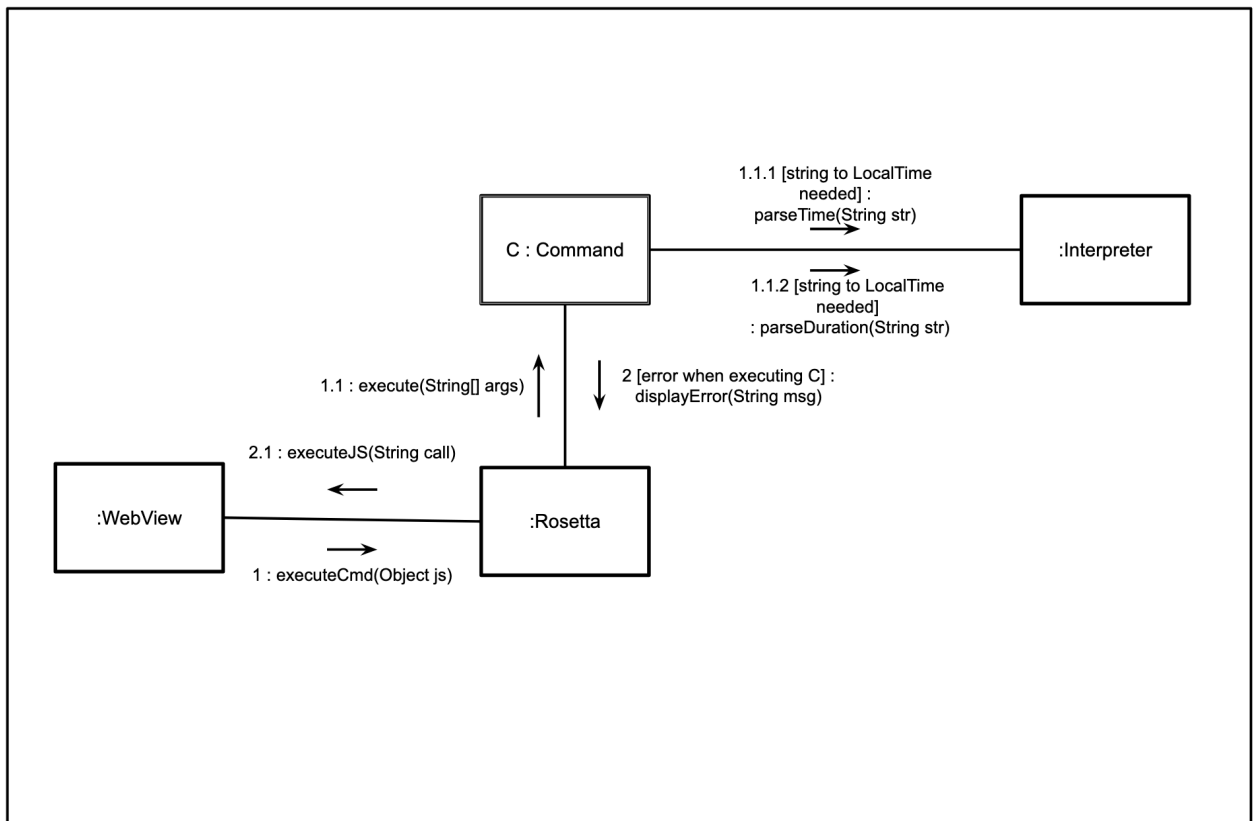


# Design

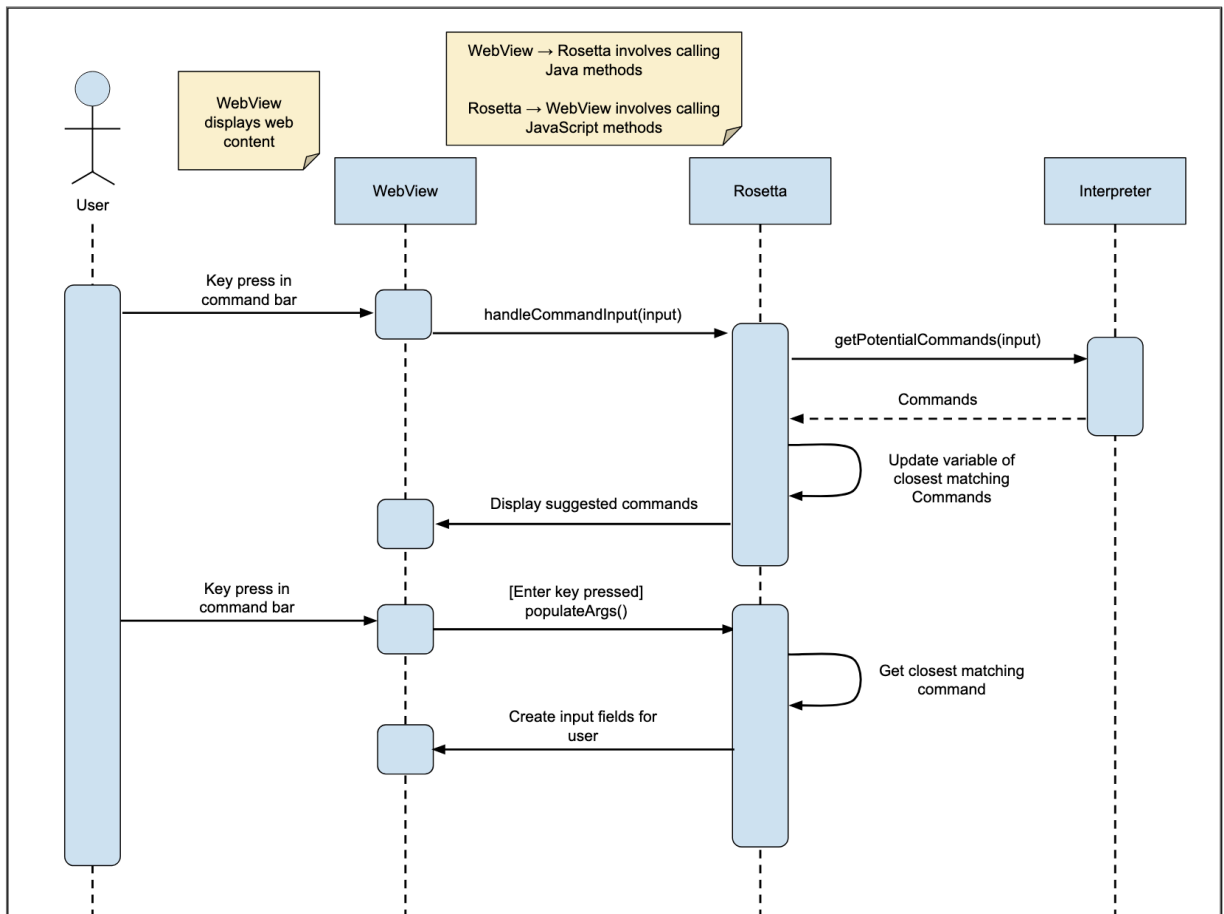
## UML Class Diagram – Java Backend Overview



## UML Communication Diagram – Frontend-Backend Bridge



## UML Sequence Diagram – Autocomplete in command bar



## User Interface Sketch – Home screen

This is the current time that is detected

12:34pm

Homework 13:00 – 14:00

Meeting 16:00 – 16:30

Dinner 19:00 – 21:00

Tasks are listed out for the current day.

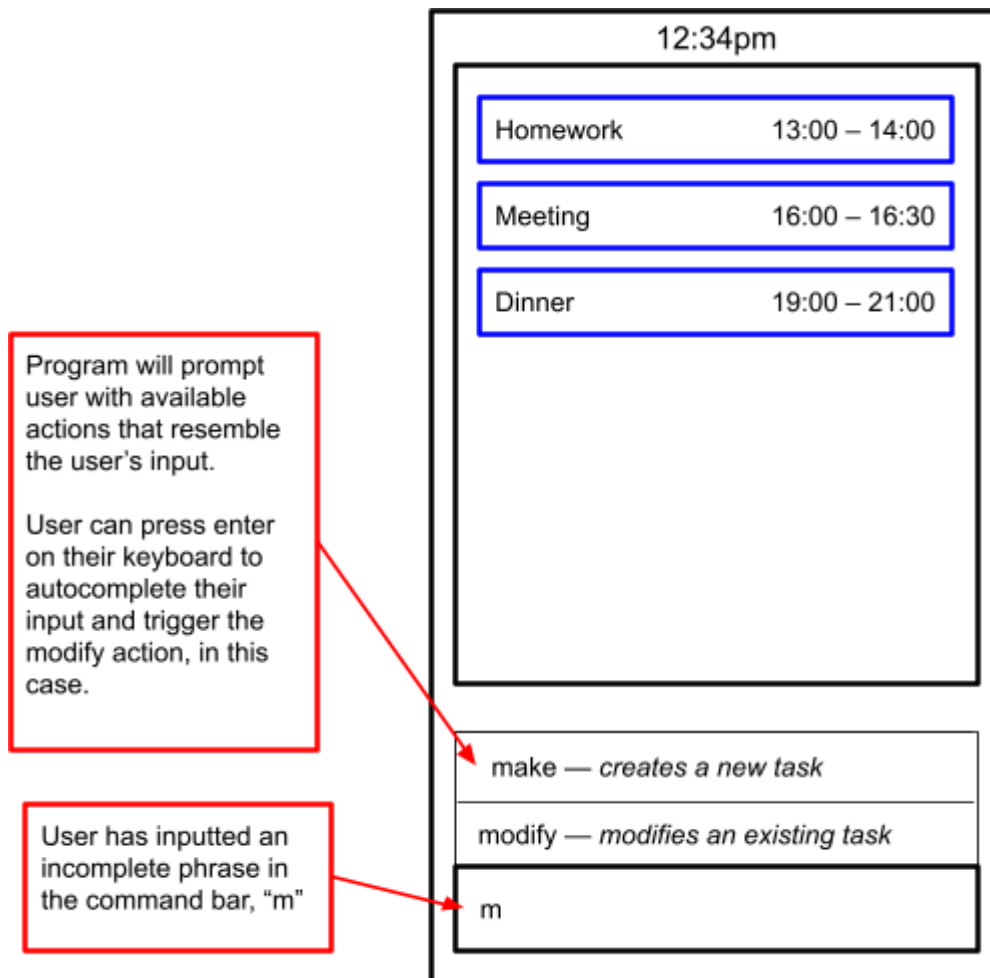
Users can type in the command bar to trigger actions to add, modify and delete tasks

*Enter a command*

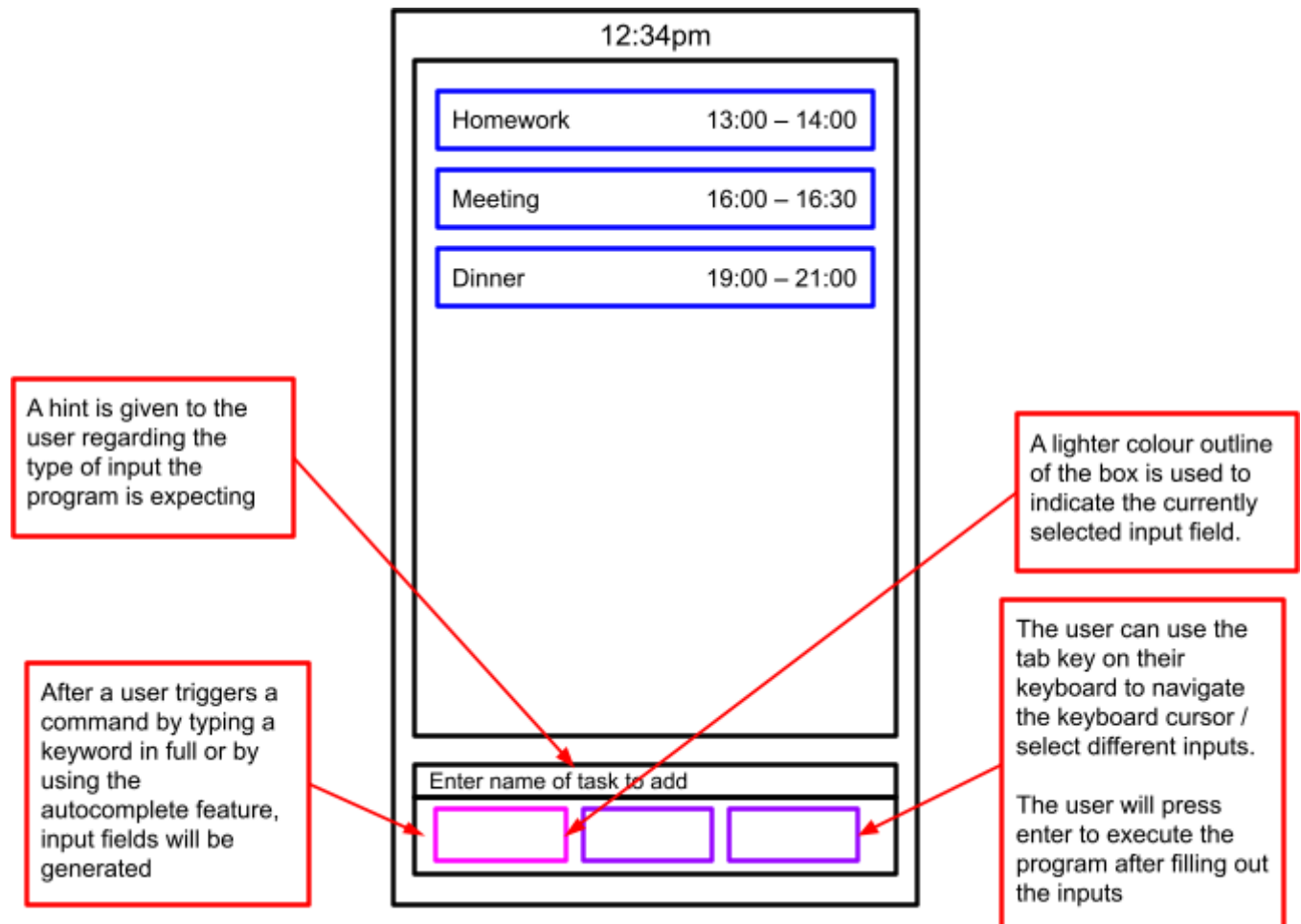
The sketch shows a mobile application interface. At the top, the current time '12:34pm' is displayed. Below it, a list of tasks for the current day is shown, each with a title and a time range: 'Homework' (13:00 – 14:00), 'Meeting' (16:00 – 16:30), and 'Dinner' (19:00 – 21:00). At the bottom, there is a command bar with the placeholder text 'Enter a command'.

| Task     | Time Range    |
|----------|---------------|
| Homework | 13:00 – 14:00 |
| Meeting  | 16:00 – 16:30 |
| Dinner   | 19:00 – 21:00 |

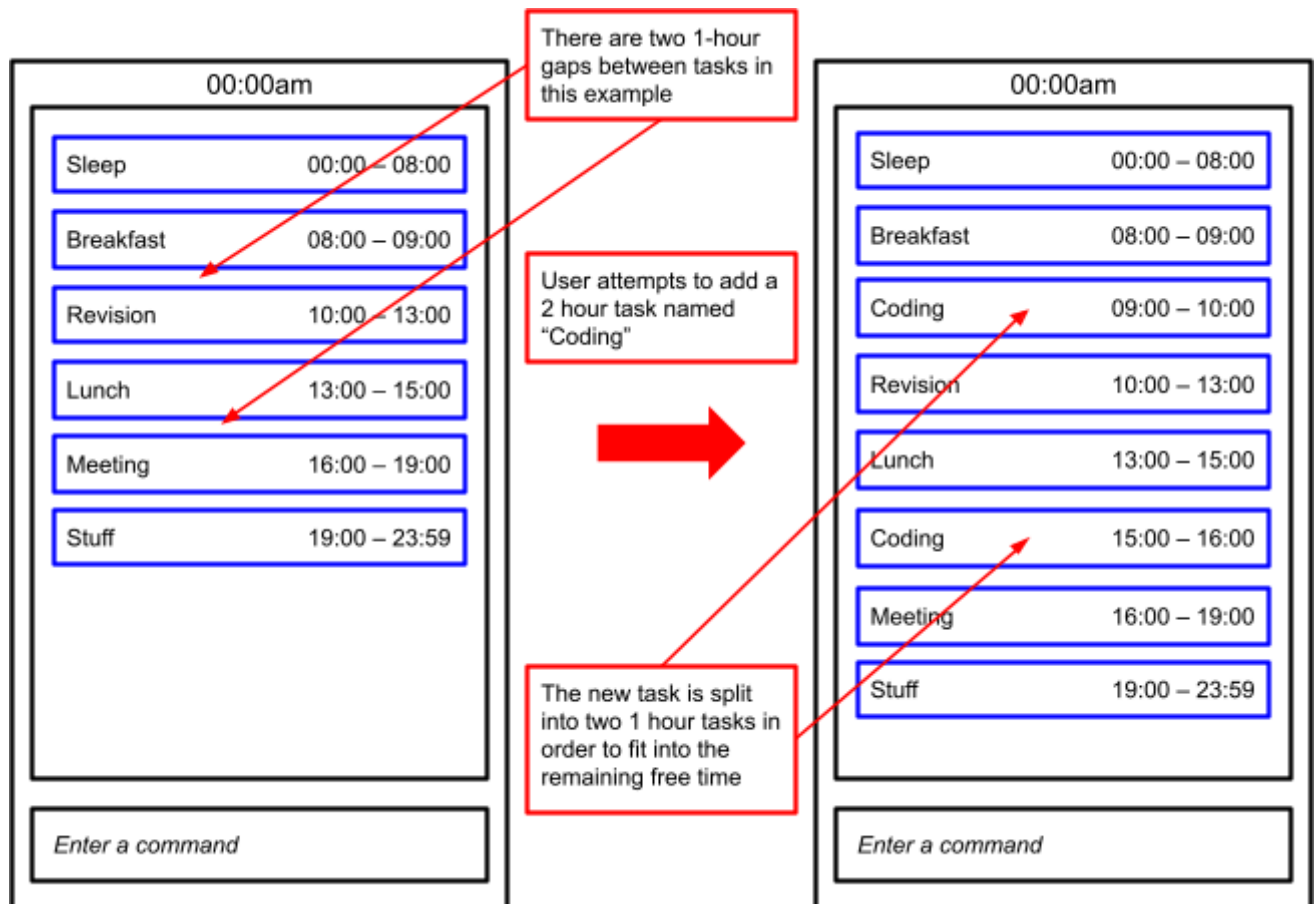
## User Interface Sketch – Inputting a command



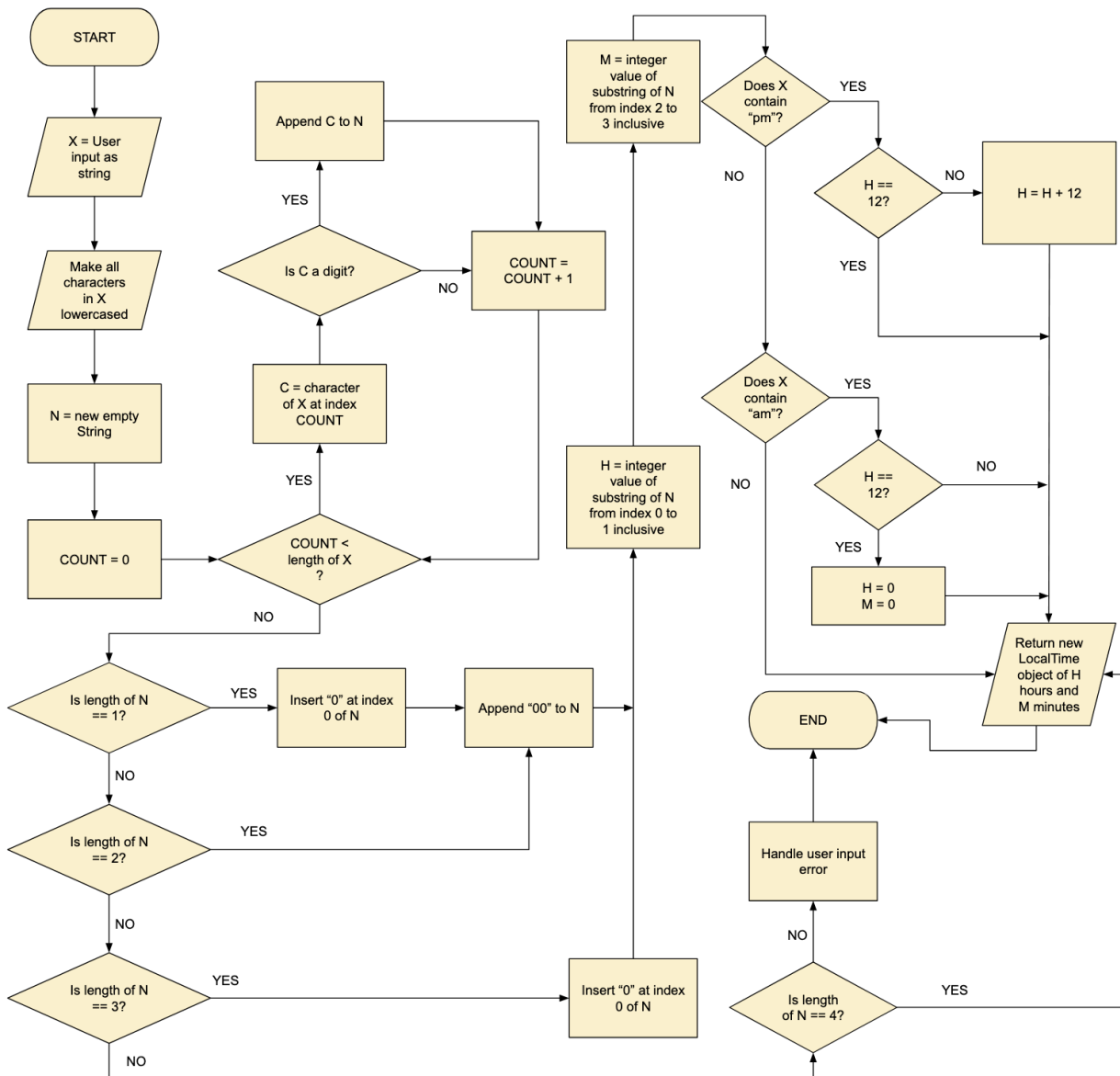
## User Interface Sketch – Input fields



## User Interface Sketch – Fragmentation of task

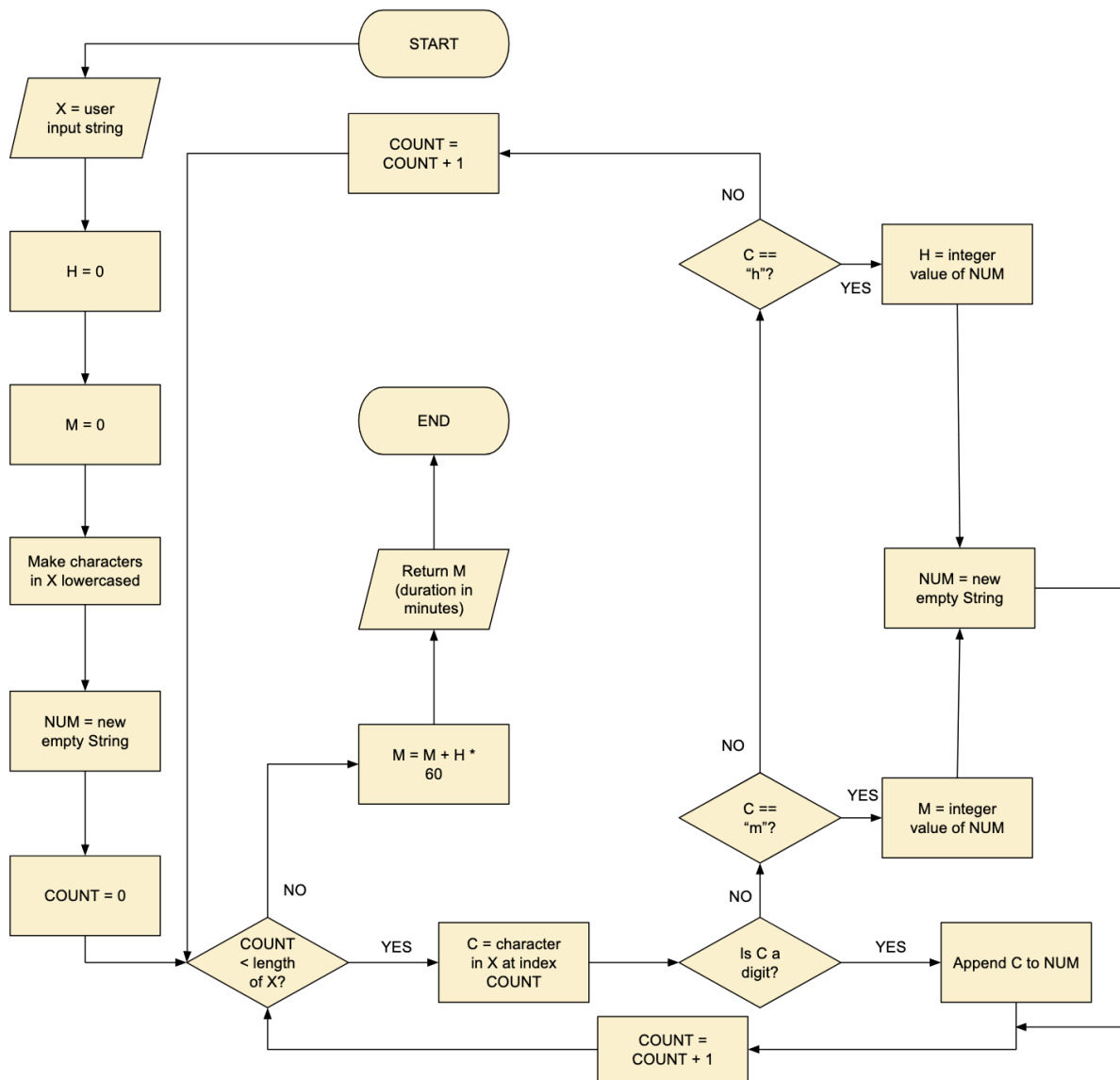


## Algorithm Flowchart – Gets time with Named Entity Recognition

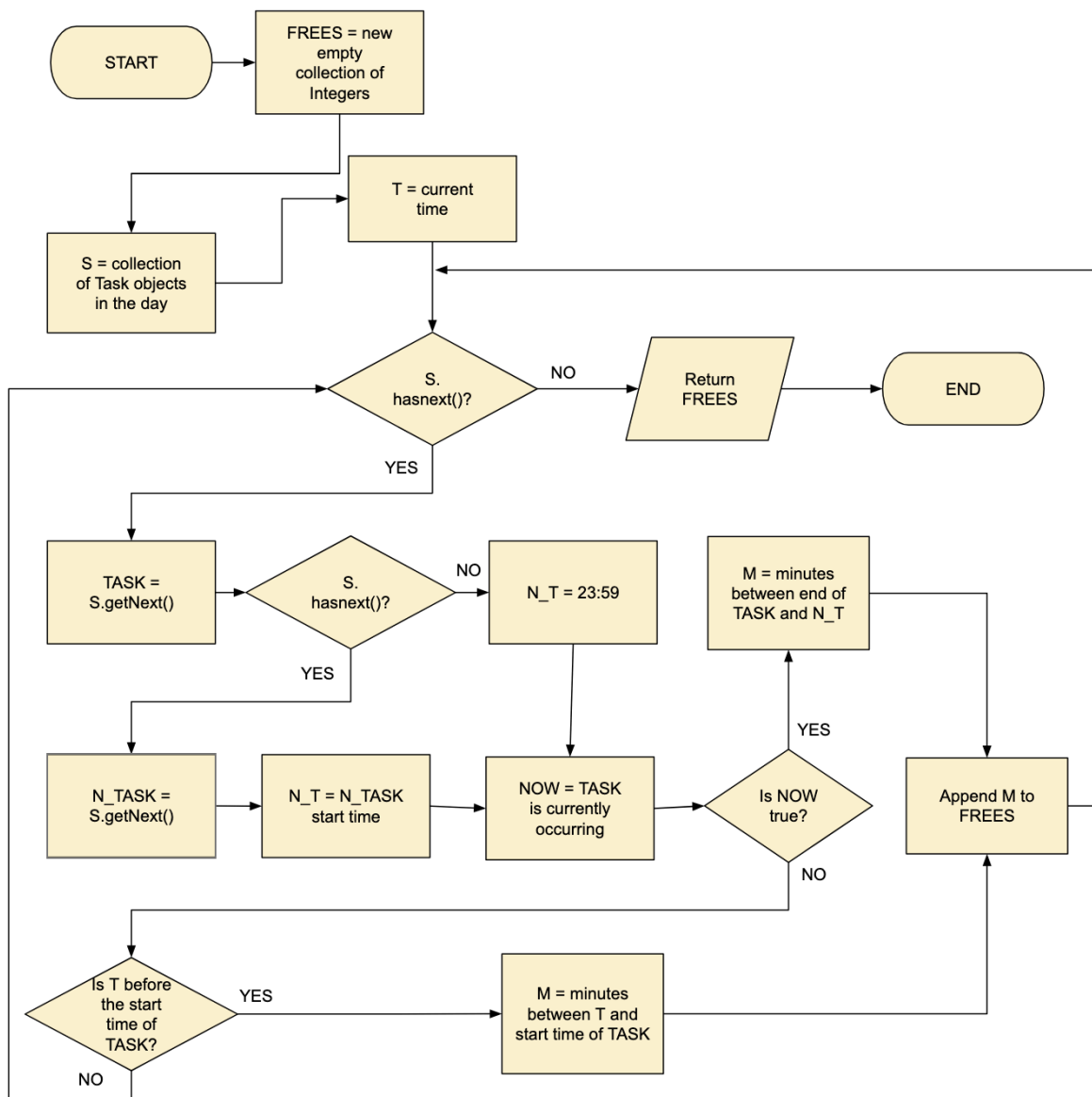




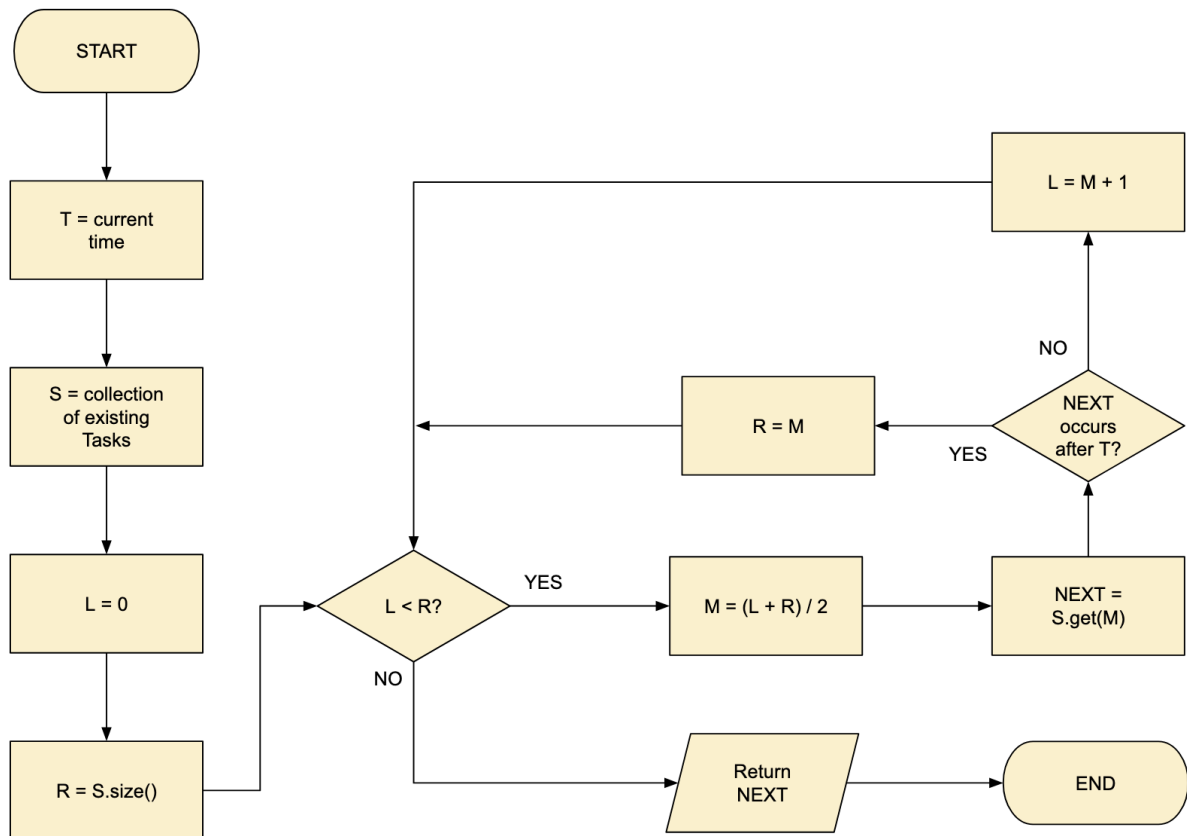
## Algorithm Flowchart – Gets duration with Named Entity Recognition



## Algorithm Flowchart – Gets free times as a collection of integers

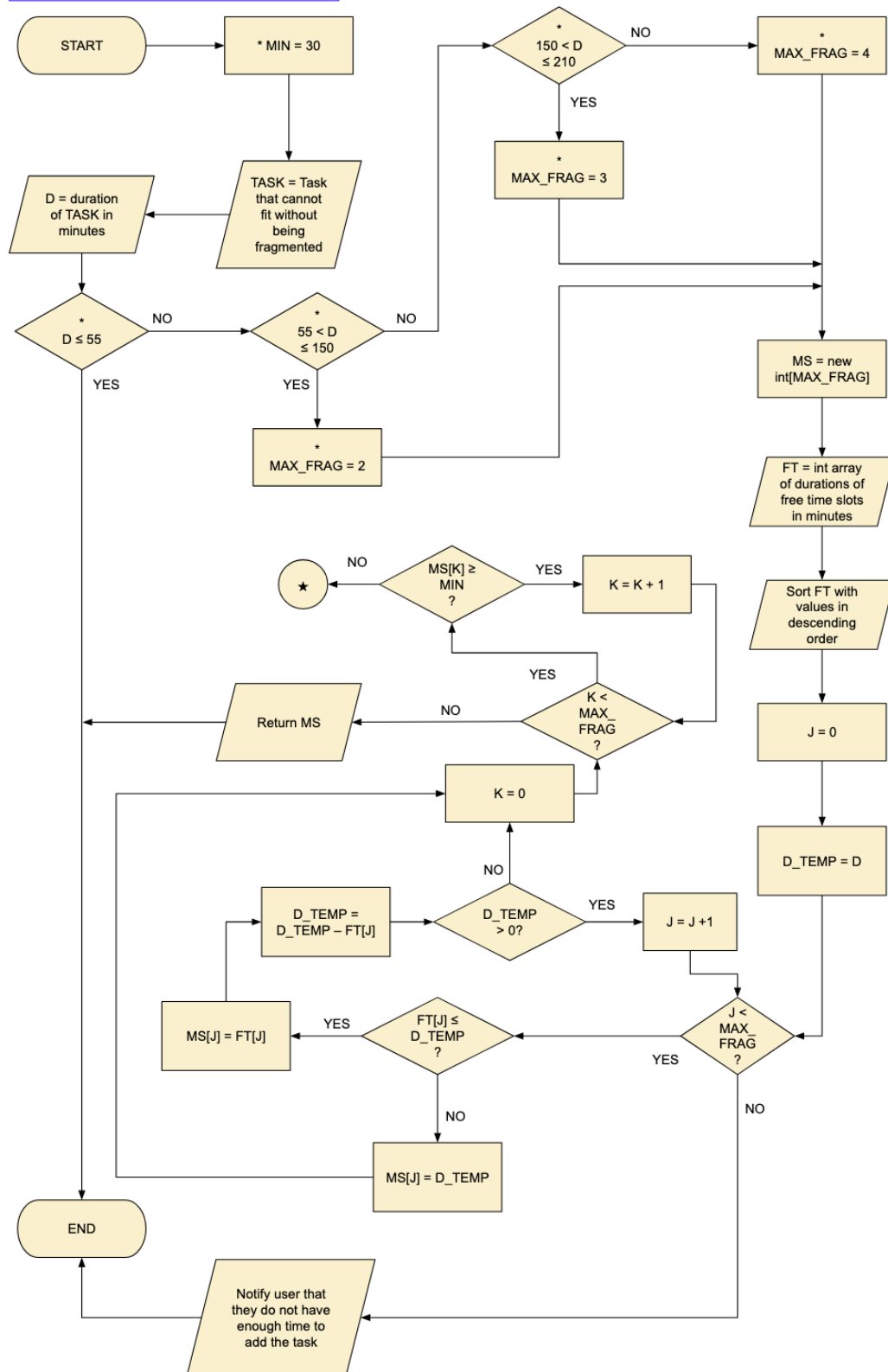


## Algorithm Flowchart – Gets the strictly next occurring task

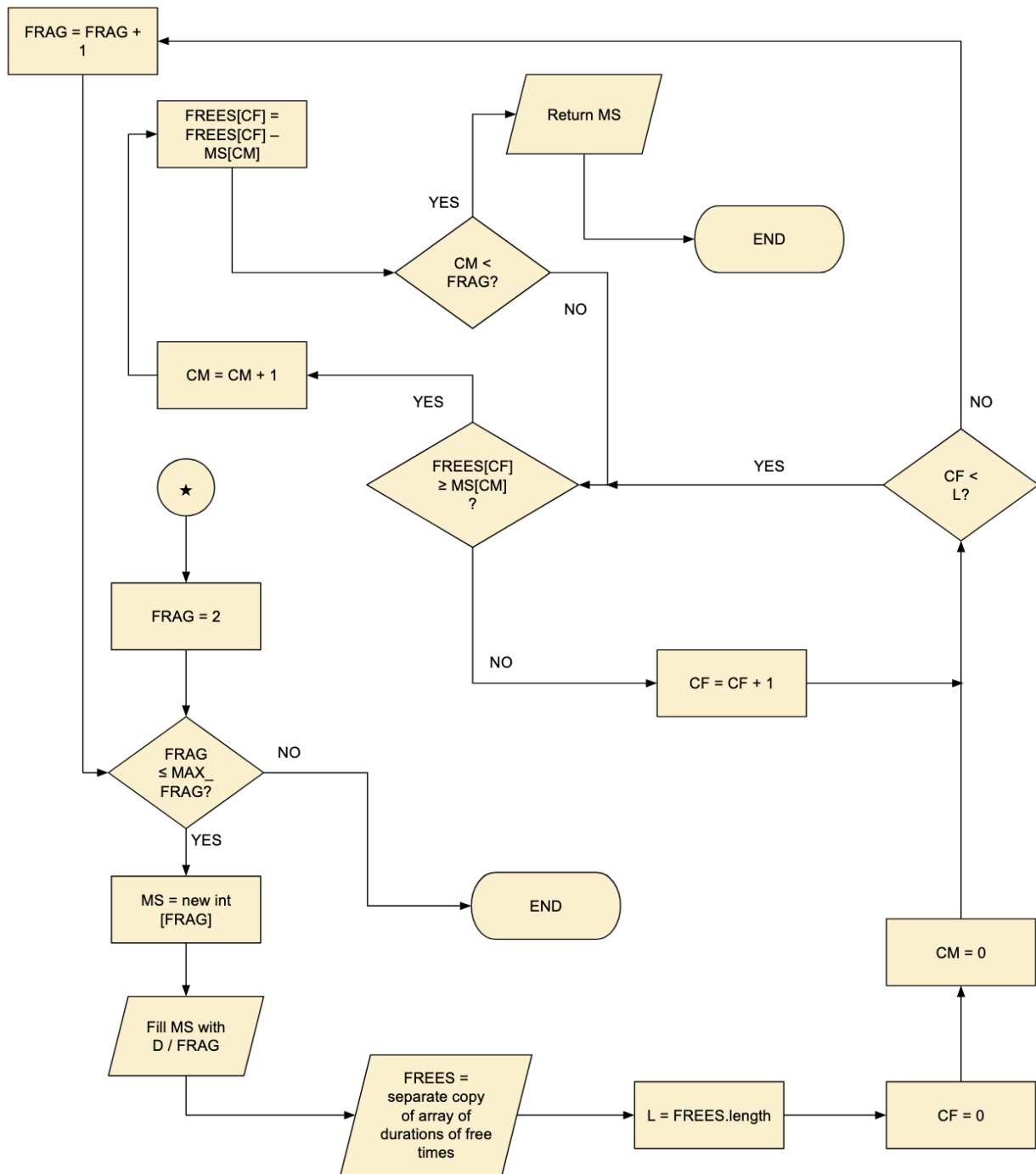


## Algorithm Flowchart – Fragments a task: Greedy

\* DESIGNED WITH REFERENCE TO APPENDIX C



## Algorithm Flowchart – Fragments a task (cont.): Even distribution



## Test plan

| Involved Success Criteria(s) | Test Description                                                         | Expected behaviour                                                                                              |
|------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1, 2                         | All program functions should be usable without a mouse                   | User can add a new task using their keyboard only                                                               |
| 3                            | Program will not require user input for a start time for a task          | Program will use the user's system time as the start time if a task does not have a specified start time        |
| 3                            | Program will not require user input for an end time for a task           | Program will calculate the end time automatically for a given start time and duration                           |
| 4                            | Program will accept a start time for a task                              | If start time is not in the past, task will be added to the day with the specified start time                   |
| 5                            | Program will split a task up into smaller tasks automatically            | Adds a 2 hour task to the day. If necessary, it will be split into two 1 hour tasks to fit between other tasks. |
| 6                            | User changes the name of a task that has been added to the day           | Changes are reflected in the GUI                                                                                |
| 6                            | User changes the start time for a task that has been added to the day    | Start time is changed, if task does not overlap other tasks                                                     |
| 6                            | User changes the task duration for a task that has been added to the day | Duration will be changed as specified unless it overlaps with other tasks                                       |
| 7                            | Name of task to be deleted is specified for deletion                     | The task is deleted                                                                                             |
| 8                            | Program should accept shorthand time inputs                              | "7pm" is inputted as a start time of a task<br><br>Task is added 19 hours from the start of the day             |

|   |                                                                                |                                                                                                             |
|---|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 8 | Program should accept time in the 24-hour format                               | <p>“2200” is inputted as a start time of a task</p> <p>Task is added 22 hours from the start of the day</p> |
| 8 | Program should accept shorthand duration inputs                                | <p>“2h” is inputted as a start time of a task</p> <p>Task is added with a duration of 2 hours</p>           |
| 8 | Program should accept minutes and hours in duration inputs                     | <p>“1h30min” is inputted as a duration of a task</p> <p>Task is added with a duration of 1.5 hours</p>      |
| 9 | Program actions should be faster than alternative solutions that are available | Program should take less time than an alternative program to complete the same set of actions               |

0 words